

PROJECT BIBLE — BT-C9C4C5 v2.4 (Technical Reference)

Format: Machine-readable / AI-LLM ingestion
Tag: BT-C9C4C5
Version: 2.4 — Trinity Protocol Upgrade
Generated: 2026-06-05
Supersedes: v2.2 (Mainnet Core), v2.3 (Macro-Asset Pipeline)
Inheritance Scope: All future modules MUST conform to this document.

§1 — SYSTEM IDENTITY

```
project_name: Bedrock ESG
project_type: Industrial Carbon Verification Platform
domain: bedrockesg.com
repo_tag: v2.4-trinity-protocol
stack: Fullstack Web Application (Server-Side Rendering)
db_engine: PostgreSQL (Prisma ORM)
blockchain_cluster: solana-mainnet-beta
notary_identity: "Aethexer Notary Node"
notary_node_id: "Sentinel Node-01"
notary_version: "2.4.0"
auth_protocol: "aethexer-v1"
rbac_version: "1.0.0"
audit_pipeline_version: "1.0.0"
sse_ticker_version: "1.0.0"
```

§2 — SOVEREIGN SETTLEMENT MODEL (70/20/10)

§2.1 — Constants

```
split:
  FOUNDER_YIELD: 0.70 # Display label: "Verified Asset Value" / "Asset Sovereign"
  STEWARDSHIP: 0.20 # Display label: "Platform Processor"
  PUBLIC_RESILIENCE: 0.10 # Display label: "Public Resilience"
  SUM_INVARIANT: 1.00 # MUST always equal 1.0 — runtime assertion enforced
```

§2.2 — Display Labels (SETTLEMENT_LABELS)

Constant Key	Percentage	Display Label	Description
SEVENTY	70%	Verified Asset Value	Direct liquidity for the Verified Asset Holder
TWENTY	20%	Platform Processor	Maintenance of Bed-rock ESG / Unicon / Freality nodes
TEN	10%	Public Resilience	Non-custodial routing to resilience grants

§2.3 — Zero Greed Policy

IF (publicResilience / total) < 0.10 - 0.001 THEN status = HIGH_FRICTION
 HIGH_FRICTION blocks settlement until ratio is restored.
 Tolerance: ±0.1% for rounding errors only.

§2.4 — Rules

- The 10% Public Resilience block is a **singular, unified block**.
 - **BANNED**: Hardcoded sub-splits within the 10% tier.
 - Dynamic routing by `assetClass` is permitted at the metadata level only (`assetClassMetadata` JSON field).
 - Internal DB field names (`founderYieldUsd` , `stewardshipUsd` , `publicResilienceUsd`) are frozen for schema compatibility.
-

§3 — ASSET CLASS VARIANTS (A-E)

```

asset_class_variants:
  A:
    code: SOVEREIGN_SKIN
    label: "Sovereign Skin™ / Gripsy"
    characteristics: Thermodynamic + Kinetic
    description: Asphalt recycling, surface materials, circular construction inputs
  B:
    code: PROPERTY_INDUSTRIAL
    label: "Commercial & Industrial Property"
    characteristics: Commercial / Industrial Real Estate
    sub_classifications:
      - VERTICAL_REAL_ESTATE      # Skyscrapers / high-rise commercial towers
      - CASINO_COMPLEX            # Entertainment / casino mega-structures
      - INDUSTRIAL_LOGISTICS      # Warehouses, distribution centres, factories
    description: Macro-asset pipeline with telemetry, spatial scanning, and discovery
  C:
    code: PROPERTY_RESIDENTIAL
    label: "Property – Residential"
    characteristics: Residential Real Estate
    description: Single-family, multi-family, condominiums, rental properties
  D:
    code: PHYSICAL_COMMODITIES
    label: "Physical Commodities"
    characteristics: Fiat Currency, Commodities
    description: Currency destruction, commodity extraction, dead mass processing
  E:
    code: PRECIOUS_METALS_CUSTODIAL
    label: "Precious Metals Custodial"
    characteristics: Gold, Silver, Platinum
    description: Custodial pipeline with assay verification

```

§3.1 — Precious Metals VAL Formula (Variant E)

$$\text{VAL} = (\text{Spot Price} \times \text{Weight} \times \text{purity_percentage}) - \text{Refinery Discount}$$

§3.2 — Variant B Sub-Classifications (Prisma Enum `AssetSubClass`)

```

VERTICAL_REAL_ESTATE:  # Skyscrapers: Burj Khalifa, Taipei 101, One WTC, Shanghai Tower
CASINO_COMPLEX:        # Casinos: MGM Grand, Venetian Macau, Marina Bay Sands
INDUSTRIAL_LOGISTICS:  # Warehouses, distribution centres, factories

```

§3.3 — Routing Rule

- Only the five codes (A-E) are valid. **BANNED**: Fabricated variant codes.
- All new Variant B assets enter at `PENDING_SOVEREIGN_REVIEW` by default.

§4 — ROLE-BASED ACCESS CONTROL (v2.4)

§4.1 — Role Hierarchy

```
roles:
  FOUNDER:
    level: 100
    description: Full system access – sovereign override
    permissions: ALL (19 permissions)
  ADMIN:
    level: 75
    description: Full platform access except founder-only settings
    exclusions: [system:full, audit_logs:export, roles:manage, founder:settings]
  SOVEREIGN_AGENT:
    level: 50
    description: Field verification, submission approval
    permissions: [submissions:approve, submissions:verify, submissions:create, submissions:view_own, submissions:view_all, settlements:view, discovery:view, telemetry:ingest, telemetry:view]
  CLIENT_OWNER:
    level: 25
    description: Submit assets, view own submissions only
    permissions: [submissions:create, submissions:view_own, telemetry:view]
```

§4.2 — Legacy Role Mapping

```
legacy_mappings:
  ADMIN: ADMIN
  AGENT: SOVEREIGN_AGENT
  FARMER: CLIENT_OWNER
  REALTOR: CLIENT_OWNER
  BUSINESS_OWNER: CLIENT_OWNER
  SOVEREIGN_INDIVIDUAL: CLIENT_OWNER
```

§4.3 — Permission Catalogue (19 permissions)

```
permissions:
- system:full           # Founder-only
- platform:manage       # Admin+
- submissions:approve   # Sovereign Agent+
- submissions:verify    # Sovereign Agent+
- submissions:create    # Client Owner+
- submissions:view_own  # Client Owner+
- submissions:view_all  # Admin+
- settlements:execute   # Admin+
- settlements:view      # Sovereign Agent+
- audit_logs:view       # Admin+
- audit_logs:export     # Founder only
- roles:manage          # Founder only
- discovery:manage      # Admin+
- discovery:view        # Sovereign Agent+
- users:manage          # Admin+
- telemetry:ingest      # Sovereign Agent+
- telemetry:view        # Client Owner+
- notary:anchor         # Admin+
- founder:settings      # Founder only
```

§4.4 — Implementation Files

```
core_engine: lib/rbac.ts           # resolveRole(), hasPermission(), roleLevel(), can-
ModifyRole()
api_guard: lib/rbac-guard.ts       # withRbac() wrapper using custom JWT auth (auth-
token cookie / Bearer)
auth_pattern: Custom JWT (not NextAuth)
jwt_secret_env: JWT_SECRET
token_location: [cookie:auth-token, header:Authorization]
```

§4.5 — Role Modification Guard

```
canModifyRole(actorRole, targetCurrentRole, targetNewRole):
  actorLevel > targetLevel AND actorLevel > newLevel
  → FOUNDER can modify anyone below FOUNDER
  → ADMIN can modify SOVEREIGN_AGENT and CLIENT_OWNER
  → No self-elevation possible
```

§5 — AUDIT LOG PIPELINE (v2.4)

§5.1 — AuditLog Model

```
model: AuditLog
table_name: audit_logs
fields:
  id: UUID (PK)
  createdAt: DateTime (auto)
  actorId: String (FK to User)
  actorRole: UserRole enum
  eventType: String
  entityType: String # Submission, Settlement, User, TelemetryReading
  entityId: String
  previousValue: Text (nullable, JSON)
  newValue: Text (nullable, JSON)
  metadata: Text (nullable, JSON)
  auditHash: String (SHA-256 tamper detection)
  backtraceId: String (default: BT-C9C4C5)
indexes:
  - actorId
  - eventType
  - [entityType, entityId] (composite)
  - createdAt
```

§5.2 — Event Types

```
event_types:
- SUBMISSION_STATUS_CHANGE
- SETTLEMENT_EXECUTED
- BLOCKCHAIN_ANCHOR
- ROLE_CHANGE
- VERDICT_OVERRIDE
- FOUNDER_APPROVAL
- TELEMETRY_INGEST
- DISCOVERY_FLAGGED
```

§5.3 — Audit Hash Formula

```
Input = JSON.stringify({
  actorId, eventType, entityId,
  previousValue || '', newValue || '', timestamp
})
Hash = SHA-256(Input) → hex string
```

§5.4 — Implementation Files

```
core_logger: lib/audit-logger.ts
convenience_helpers:
- logSubmissionStatusChange(actorId, actorRole, submissionId, prevStatus, newStatus,
  metadata?)
- logSettlementExecution(actorId, actorRole, settlementId, splitData, metadata?)
- logBlockchainAnchor(actorId, actorRole, entityType, entityId, txHash, metadata?)
- logRoleChange(actorId, actorRole, targetUserId, prevRole, newRole)
- logVerdictOverride(actorId, actorRole, entityType, entityId, prevVerdict,
  newVerdict, note?)
query_endpoint: GET /api/audit-logs (ADMIN+ via withRbac)
pattern: Fire-and-forget (errors logged, never break caller)
```

§5.5 — Wired Integration Points

```
current_integrations:
- /api/submissions/[id]/settle → logSettlementExecution() after settlement creation
future_targets:
- /api/v1/submissions/[id]/approve → logSubmissionStatusChange()
- /api/v1/telemetry-ingest → logBlockchainAnchor()
- /api/auth/* → logRoleChange() on role mutations
```

§6 — 7-TIER SENTINEL VERDICT MATRIX

§6.1 — Composite Score Formula

```
Composite = (docScore × 0.40) + (llmScore × 0.35) + (spatialScore × 0.25)
```

Weight	Dimension	Threshold	Method
40%	Document Integrity	≥95%	SHA-256 hash + magic byte validation
35%	LLM Parser Confidence	≥75%	AI extraction quality scoring
25%	Spatial Correlation	≥75%	Geolocation + metadata coherence

§6.2 — Verdict Tiers (Prisma SubmissionStatus + TelemetryReading.sentinelVerdict)

Verdict	Trigger Condition	Action
AUTO_APPROVED	≥95% composite / 0 failures	Settlement triggered
PENDING_SOVEREIGN_REVIEW	75-94% composite / ≤1 failure	Manual review queue
PENDING_AUTHORIZATION	65-74% composite / ≤1 failure	Escrow hold
SOVEREIGN_HOLD	Manual trigger	Perimeter flag
ESCROW_REVIEW	Condition-based	Settlement hold
PENDING_PLATFORM_REVIEW	Structural issue	Infrastructure validation
REJECTED	<65% composite OR ≥2 failures	No settlement

§6.3 — Discovery Tier (Prisma Enum DiscoveryTier)

tiers:

- PENDING_SOVEREIGN_REVIEW # Default for all new unlisted assets
- AUTO_APPROVED # After sovereign review confirms the asset

§7 — SSE MARKET LISTENER (v2.4)

§7.1 — Architecture

```

endpoint: /api/ticker-stream
protocol: Server-Sent Events (SSE)
interval: 3–5 seconds (randomized)
fallback: Client-side local jitter simulation if SSE connection fails
pattern: ReadableStream with TextEncoder

```

§7.2 — Event Types

```

events:
  ticker:
    description: Full asset matrix update
    payload:
      assets: Array<{ id, symbol, name, category, baseValue, unit, verdict,
sub_classification?, value, change, changePercent, lastUpdated }>
      humBaselineMw: 12.6
      co2eDaily: 74088 # tonnes CO2e/day across tracked exchanges
  hum_anchor:
    description: HUM frequency shift detected (≥5% from baseline)
    payload:
      assetId: string
      shiftPercent: number
      baselineMw: 12.6
      currentMw: number
      timestamp: ISO-8601

```

§7.3 — 12 Seeded Macro-Assets (Canonical Symbols)

```
exchanges:
- { id: NYSE-01, symbol: NYSE-01, name: "New York Stock Exchange", category: EX-
CHANGE, unit: MW, verdict: AUTO_APPROVED }
- { id: NASDAQ-01, symbol: NASDAQ-01, name: "Nasdaq Composite", category: EXCHANGE,
unit: MW, verdict: AUTO_APPROVED }
- { id: LSE-01, symbol: LSE-01, name: "London Stock Exchange", category: EXCHANGE,
unit: MW, verdict: AUTO_APPROVED }
- { id: CME-01, symbol: CME-01, name: "Chicago Mercantile Exchange", category: EX-
CHANGE, unit: MW, verdict: AUTO_APPROVED }
- { id: EUREX-01, symbol: EUREX-01, name: "Eurex Exchange", category: EXCHANGE,
unit: MW, verdict: AUTO_APPROVED }
skyscrapers:
- { id: BURJ-DXB-001, symbol: BURJ-DXB-001, name: "Burj Khalifa – Dubai", category:
SKYSCRAPER, unit: kW, verdict: PENDING_SOVEREIGN_REVIEW, sub: VERTICAL_REAL_ESTATE }
- { id: TAI-101-TPE, symbol: TAI-101-TPE, name: "Taipei 101 – Taiwan", category: SKY
SCRAPER, unit: kW, verdict: PENDING_SOVEREIGN_REVIEW, sub: VERTICAL_REAL_ESTATE }
- { id: OWT-NYC-001, symbol: OWT-NYC-001, name: "One World Trade – NYC", category: SKY
SCRAPER, unit: kW, verdict: PENDING_SOVEREIGN_REVIEW, sub: VERTICAL_REAL_ESTATE }
- { id: SHA-TWR-001, symbol: SHA-TWR-001, name: "Shanghai Tower – PRC", category: SKY
SCRAPER, unit: kW, verdict: PENDING_SOVEREIGN_REVIEW, sub: VERTICAL_REAL_ESTATE }
casinos:
- { id: MGM-LV-01, symbol: MGM-LV-01, name: "MGM Grand – Las Vegas", category: CASIN
O, unit: kW, verdict: PENDING_SOVEREIGN_REVIEW, sub: CASINO_COMPLEX }
- { id: VEN-MAC-01, symbol: VEN-MAC-01, name: "The Venetian – Macau", category: CASI
NO, unit: kW, verdict: PENDING_SOVEREIGN_REVIEW, sub: CASINO_COMPLEX }
- { id: MBS-SGP-01, symbol: MBS-SGP-01, name: "Marina Bay Sands – Singapore", cat-
egory: CASINO, unit: kW, verdict: PENDING_SOVEREIGN_REVIEW, sub: CASINO_COMPLEX }
```

§7.4 — Cockpit Ticker UI

```
component: components/cockpit-ticker.tsx
position: Top of main dashboard (app/page.tsx)
features:
- Auto-scrolling ticker strip with seamless loop
- SSE-driven updates (fallback: local jitter at 2.5s)
- Filter tabs: [ ALL ] | [ EXCHANGES ] | [ SKYSCRAPERS ] | [ CASINOS ]
- Verdict badges: AUTO_APPROVED (emerald), PENDING_SOVEREIGN_REVIEW (amber),
REJECTED (red)
- Flashing delta tick: cyan ring pulse (600ms) on value change
- HUM anchor indicator: ⚡ in header bar on shift events
- Version tag: BT-C9C4C5 v2.4
```

§8 — MACRO-ASSET PASSIVE DISCOVERY INTERFACE (v2.4)

§8.1 — Route & Components

```
page: /discovery (app/discovery/page.tsx)
component: components/discovery-dashboard.tsx
sidebar_section: "Discovery" (violet theme, Eye icon)
```

§8.2 — Features

capabilities:

- Passive intelligence feed for unlisted macro-assets
- 12 seeded assets with estimated values
- 70% sovereign share calculation per asset
- "RIGHT SHARE PENDING" badge (pulsing amber) for PENDING_SOVEREIGN_REVIEW
- **Filter tabs:** [ALL] | [PENDING REVIEW] | [APPROVED]
- Trigger Outreach workflow button (per pending asset)
- **Stats cards:** Auto Approved count, Pending Review count, Total Assets
- Total 70% Sovereign Value aggregate display

§8.3 — Seeded Discovery Asset Values

```
# All values in USD
exchanges:
  NYSE-01: { total: 26B, sovereign70: 18.2B, verdict: AUTO_APPROVED }
  NASDAQ-01: { total: 19.4B, sovereign70: 13.58B, verdict: AUTO_APPROVED }
  LSE-01: { total: 12.8B, sovereign70: 8.96B, verdict: AUTO_APPROVED }
  CME-01: { total: 8.2B, sovereign70: 5.74B, verdict: AUTO_APPROVED }
  EUREX-01: { total: 6.1B, sovereign70: 4.27B, verdict: AUTO_APPROVED }
skyscrapers:
  BURJ-DXB-001: { total: 1.5B, sovereign70: 1.05B, verdict: PENDING }
  TAI-101-TPE: { total: 1.8B, sovereign70: 1.26B, verdict: PENDING }
  OWT-NYC-001: { total: 3.9B, sovereign70: 2.73B, verdict: PENDING }
  SHA-TWR-001: { total: 2.4B, sovereign70: 1.68B, verdict: PENDING }
casinos:
  MGM-LV-01: { total: 4.6B, sovereign70: 3.22B, verdict: PENDING }
  VEN-MAC-01: { total: 5.2B, sovereign70: 3.64B, verdict: PENDING }
  MBS-SGP-01: { total: 5.7B, sovereign70: 3.99B, verdict: PENDING }
```

§9 — SOLANA MAINNET NOTARY

§9.1 — Configuration

```
memo_program_id: "MemoSq4gqABAXKb96qnH8TysNcWxMyWCqXgDLGmfCHr"
expected_public_key: "9tsq8qB4P9uPRSHrjXLaDQJd93nuo4WEEtkR1tAJRAQ3"
rpc_env_var: SOLANA_MAINNET_RPC_URL
rpc_fallback: "https://api.mainnet-beta.solana.com"
secret_key_env_var: SOLANA_NOTARY_SECRET_KEY
secret_key_formats: ["base58", "JSON byte array"]
cluster: mainnet-beta
engine_name: "Sovereign Notary Bridge"
engine_version: "2.4.0"
```

§9.2 — Memo Payload Schema

```
{
  "v": "2.4",
  "sys": "BEDROCK_ESG",
  "signer": "Aethexer Notary Node",
  "node": "Sentinel Node-01",
  "t": "<CARBON_AUDIT|TRADE_SETTLEMENT|HUM_FREQUENCY_SHIFT|SPATIAL_ATTESTATION|BIOCHAR_CDR|METAL_RECOVERY>",
  "id": "<entityId>",
  "h": "<first 32 chars of dataHash>",
  "co2t": "<optional: carbonTonnes>",
  "usd": "<optional: valuationUsd>",
  "bt": "<optional: backtrace ID>",
  "ts": "<unix epoch seconds>"
}
```

§9.3 — Identity Lock

- `signer` : “**Aethexer Notary Node**” — permanent. No personal names.
- `node` : “**Sentinel Node-01**” — permanent.
- Local cockpit moniker “Zer0” is display-only; never appears on-chain.

§10 — AETHEXER PROTOCOL AUTH (Trinity 3-Layer)

§10.1 — Headers

Header	Purpose	Required	Constraint
x-aethexer-key	Protocol key (server)	One of two	Timing-safe compare
x-aethexer-field-key	Field app key (mobile/edge)	One of two	Timing-safe compare
x-aethexer-ts	Unix epoch timestamp	Optional	±300s window
x-aethexer-nonce	Replay protection token	Optional	16-128 chars, single-use

§10.2 — Nonce Lifecycle

- Stored in `IngestNonce` model (Prisma)
- TTL: 24 hours, auto-cleanup on each auth call
- Replay → HTTP 409 `NONCE_REPLAY`

§11 — VALIDATOR ENGINE (lib/validator.ts)

```

schemas:
  AssetClassVariantSchema: enum [SOVEREIGN_SKIN, PROPERTY_INDUSTRIAL, PROP-
    ERTY_RESIDENTIAL, PHYSICAL_COMMODITIES, PRECIOUS_METALS_CUSTODIAL]
  VariantBSubClassificationSchema: enum [VERTICAL_REAL_ESTATE, CASINO_COMPLEX, INDUS-
    TRIAL_LOGISTICS]
  SentinelVerdictSchema: enum [AUTO_APPROVED, PENDING_SOVEREIGN_REVIEW,
    PENDING_AUTHORIZATION, SOVEREIGN_HOLD, ESCROW_REVIEW, PENDING_PLATFORM_REVIEW, REJEC-
    TED]
  SpatialDataEnvelopeSchema: { has_geometry_coords, scan_format, mesh_density_points,
    location_anchor }
  MacroAssetMetadataSchema:
    sub_classification: VariantBSubClassificationSchema (OPTIONAL in v2.4)
    chiller_exhaust_temp: number (optional)
    square_footage_volume: number (optional)
    damper_mechanical_load_kw: number (optional)
    spatial_data_envelope: SpatialDataEnvelopeSchema (optional)
  TelemetryIngestPayloadSchema: { node_id, power_usage_kw, pue_ratio, car-
    bon_per_trade_g, grid_intensity_score, asset_class, asset_class_metadata?, macro_asset
    _metadata? }
  DiscoveryAssetSchema:
    asset_id, asset_class, sub_classification?, display_name, location_label
    initial_verdict: default PENDING_SOVEREIGN_REVIEW
    discovery_tier: enum [PENDING_SOVEREIGN_REVIEW, AUTO_APPROVED] (v2.4)
    metadata?: MacroAssetMetadataSchema

```

§12 — SCHEMA CHANGES (v2.4 Delta)

§12.1 — New Enum Values

```

UserRole (extended):
  new: [FOUNDER, SOVEREIGN_AGENT, CLIENT_OWNER]
  existing: [ADMIN, AGENT, FARMER, REALTOR, BUSINESS_OWNER, SOVEREIGN_INDIVIDUAL]

SubmissionStatus (extended):
  new: [AUTO_APPROVED, PENDING_SOVEREIGN_REVIEW, PENDING_AUTHORIZATION,
    SOVEREIGN_HOLD, ESCROW_REVIEW, PENDING_PLATFORM_REVIEW, REJECTED]
  existing: [DRAFT, PENDING_FORENSIC_VERIFICATION, FORENSIC_VERIFIED, SETTLE-
    MENT_COMPLETE, REJECTED_FORENSIC]

```

§12.2 — New Enums

```

DiscoveryTier: [PENDING_SOVEREIGN_REVIEW, AUTO_APPROVED]
AssetSubClass: [VERTICAL_REAL_ESTATE, CASINO_COMPLEX, INDUSTRIAL_LOGISTICS]

```

§12.3 — New Model: AuditLog

(See §5.1 for full specification)

§12.4 — Migration Strategy

```
method: prisma db push (non-destructive)
no_data_loss: true
migrate_deploy: NOT executed – left for manual admin trigger
```

§13 — SUBMISSION TRACKS

```
tracks:
  TRACK_A_MATERIALS:
    label: "Materials"
    scope: Asphalt, Concrete, Metal
    coefficient_source: ICE v4.1
  TRACK_B_ENERGY_179D:
    label: "179D Energy"
    scope: Building envelopes, energy audits
    compliance: IRS Form 7205
  TRACK_C_BIOCHAR_CDR:
    label: "Biochar CDR"
    scope: Biochar production receipts
    output: CDR credit linkage
```

§14 — SUBMISSION STATE MACHINE (v2.4 Extended)

```
DRAFT
→ PENDING_FORENSIC_VERIFICATION
  → FORENSIC_VERIFIED
    → [founderApproved = true]
      → SETTLEMENT_COMPLETE

→ AUTO_APPROVED (via Sentinel ≥95% composite)
  → SETTLEMENT_COMPLETE

→ PENDING_SOVEREIGN_REVIEW (via Sentinel 75-94%)
  → [sovereign review] → AUTO_APPROVED or REJECTED

→ PENDING_AUTHORIZATION (65-74%)
→ SOVEREIGN_HOLD (manual)
→ ESCROW_REVIEW (conditional)
→ PENDING_PLATFORM_REVIEW (structural)
→ REJECTED (<65% or ≥2 failures)
→ REJECTED_FORENSIC (legacy forensic failure)
```

§15 — EXITZ TOKEN ECOSYSTEM (Solana Mainnet)

```
tokens:
  EXITZ_BASE:
    mint: "44SrHT9QwyZ2jiiJKRF1zHKF2NFTgTC8m6ubPTb5qJBJ"
    supply: 2
  EXITZ_C:
    mint: "7ZPsJE3W5YXnnYgapzfsA5JxtK8VQwtTn5Jb23hgFzMK"
    supply: 1
    description: Classic Vehicles
  EXITZ_A:
    mint: "G11hBknKcBXUjc4kvAtAkWjoHz5yMr7FTqEEYC96EnJn"
    supply: 1
    description: Fine Art
program: Token-2022
mobile_assets: VIN/Odometer integrity (no HUM)
stationary_assets: HUM Seismic Baseline
```

§16 — HUM LISTENER

```
baseline_mw: 12.6
shift_threshold: 5% (env: HUM_TRIGGER_THRESHOLD, default 0.5% for high-freq testing)
co2e_daily: 74,088 tonnes across tracked exchanges
sources: [NYSE/NASDAQ, LSE, CME/Eurex]
sse_integration: /api/ticker-stream emits hum_anchor events on threshold breach
api_routes:
  - GET /api/hum-telemetry (state)
  - POST /api/hum-telemetry (read)
```

§17 — CARBON CALCULATION ENGINES

```
asphalt:
  api: /api/calculate
  method: Eurobitume A1-A3 LCA (local engine, ZERO external calls)
  baseline: 60 kg CO2/tonne (virgin)
  rap_reduction: 0.4 kg/tonne per %RAP
biochar:
  api: /api/biochar
  framework: Hedera Guardian
  attestation: Dual-chain (Hedera HCS + Polygon ERC-1155)
metal:
  api: /api/metal
  supported: [STEEL, ALUMINUM, COPPER, BRASS, LEAD, ZINC, STAINLESS_STEEL, CAST_IRON,
TITANIUM, NICKEL]
  factors: ICE Database standard
```

§18 — EMISSION FACTOR DATABASE

```
sources:
  - ICE Database v4.1 (25 factors)
  - Ecoinvent v3.10.1 (15 factors)
total: 40
biogenic_correction: 60% (US, BR markets)
grid_mixes:
  US: 0.419, BR: 0.082, EU: 0.253, GB: 0.207 kgCO2e/kWh
api: /api/emission-factors
```

§19 — MAGIC BYTE VALIDATION

MIME Type	Magic Bytes	Description
application/pdf	25 50 44 46	%PDF
image/jpeg	FF D8 FF	JPEG
image/png	89 50 4E 47	.PNG
XLSX/DOCX	50 4B 03 04	PK (ZIP)
PLY (spatial)	"ply\n" ASCII header	PLY 3D scan
Splat (spatial)	≥32 bytes	Gaussian Splat

Mismatch → HTTP 422 `MAGIC_BYTE_MISMATCH`.

§20 — DATABASE GUARDRAILS

```
max_concurrent_connections: 25
idle_session_timeout: short
statement_timeout: 5s
idle_in_transaction_timeout: 30s
connection_model: ephemeral
schema_migration: prisma db push (non-destructive only)
banned: prisma migrate deploy, prisma db push --accept-data-loss, prisma db push --force-reset
```

§21 — API ROUTE MAP (v2.4 Complete)

Route	Method	Purpose	RBAC Gate
/api/calculate	POST	Asphalt carbon LCA	—
/api/biochar	GET/POST	Biochar CDR batches	—
/api/biochar/history	GET	Historical batch query	—
/api/biochar/stats	GET	Aggregate biochar stats	—
/api/biochar/[id]/at-testation	GET	Blockchain attestation details	—
/api/metal	GET/POST	Metal recovery calculations	—
/api/carbon-impact	GET	Unified carbon impact summary	—
/api/stats	GET	Dashboard summary metrics	—
/api/analytics	GET	Time series chart data	—
/api/projects	GET/POST	Project management	—
/api/locations	GET/POST	Location CRUD	—
/api/locations/[id]	ALL	Location by ID	—
/api/farmers	GET/POST	Farmer management	—
/api/farmers/[id]	ALL	Farmer by ID	—
/api/credits	GET/POST	Farmer credit life-cycle	—
/api/credits/stats	GET	Credit aggregate stats	—
/api/notary	GET/POST	Solana notary status/anchor	—
/api/hum-telemetry	GET/POST	HUM listener state/read	—
/api/emission-factors	GET		—

Route	Method	Purpose	RBAC Gate
		Emission factor look-up	
/api/spatial	GET/POST	Spatial PLY parser	—
/api/submissions	GET/POST	Business submissions	—
/api/submissions/[id]/parse	POST	AI forensic parse	—
/api/submissions/[id]/settle	POST	Settlement execution + audit log	—
/api/submissions/[id]/certificate	POST	Sovereign certificate PDF	—
/api/upload	POST	S3 presigned URL generation	—
/api/ticker-stream	GET	SSE live asset telemetry (v2.4)	—
/api/audit-logs	GET	Audit trail query (v2.4)	audit_logs:view
/api/consent	GET/POST	User consent management	—
/api/consent/verify	POST	Consent verification	—
/api/consent/logs	GET	Consent audit logs	—
/api/auth/login	POST	JWT authentication	—
/api/auth/register	POST	User registration	—
/api/auth/agent-signup	POST	Agent signup with invitation	—
/api/invitations	GET/POST	Invitation code management	—
/api/agent/register-farmer	POST	Agent registers farmer	—
/api/agent/pending-counts	GET	Pending verification counts	—

Route	Method	Purpose	RBAC Gate
/api/agent/verification-queue	GET/POST	Verification queue management	—
/api/v1/forensic-ingest	POST	Unified forensic ingest	Aethexer Auth
/api/v1/telemetry-ingest	POST	Telemetry ingest (Aethexer)	Aethexer Auth
/api/v1/submissions/[id]/approve	PATCH	Sovereign approval gate	JWT

§22 — ENVIRONMENT VARIABLES (Required)

```
# Solana Notary
SOLANA_NOTARY_SECRET_KEY: <base58 or JSON array>
SOLANA_MAINNET_RPC_URL: <RPC endpoint>

# Aethexer Protocol
AETHEXER_PROTOCOL_KEY: <server key>
AETHEXER_FIELD_KEY: <field app key>

# HUM Listener
HUM_TRIGGER_THRESHOLD: <default 0.5, production 5.0>

# Database
DATABASE_URL: <PostgreSQL connection string>

# Auth
JWT_SECRET: <JWT signing secret>
NEXTAUTH_SECRET: <auto-configured>
NEXTAUTH_URL: <auto-configured per environment>
```

§23 — BANNED PATTERNS

```
banned:
- Personal names in codebase or on-chain memos
- Operational monikers in identity fields ("Zer0" never on-chain)
- Hardcoded sub-splits within the 10% Public Resilience tier
- Fabricated node IDs (only "Sentinel Node-01")
- Fabricated asset class codes (only A-E)
- External LLM API calls from /api/calculate
- Devnet cluster references (mainnet-only)
- prisma migrate deploy without explicit admin approval
- prisma db push --accept-data-loss without explicit user confirmation
- Self-elevation of roles (canModifyRole() guard enforced)
```

§24 — INHERITANCE DIRECTIVE

All future modules added to Bedrock ESG MUST:

1. Apply the 70/20/10 split via `UNIVERSAL_SPLIT` constants from `lib/universal-law.ts`.
 2. Use `SETTLEMENT_LABELS` for all user-facing split references.
 3. Route through one of the 5 defined `ASSET_CLASS_VARIANTS` (A-E). No new variants without explicit protocol amendment.
 4. Pass Sentinel scoring before settlement. No bypass paths.
 5. Anchor all settlements to Solana Mainnet via `anchorToChain()` from `lib/solana-notary.ts`.
 6. Use "Aethexer Notary Node" as the signer identity. No personal names.
 7. Authenticate external ingest via `validateAethexerKey()` from `lib/aethexer-auth.ts`.
 8. Validate all uploaded file buffers via magic byte checks.
 9. Enforce Zero Greed Policy via `verifyZeroGreedPolicy()` before any settlement.
 10. Respect the DB connection ceiling of 25 concurrent connections.
 11. **v2.4**: Gate protected routes via `withRbac()` from `lib/rbac-guard.ts` with appropriate permission.
 12. **v2.4**: Write audit log entries via `lib/audit-logger.ts` for all state-changing operations.
 13. **v2.4**: New Variant B assets MUST enter at `PENDING_SOVEREIGN_REVIEW`.
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